VULTR AGENT RUSH · CHALLENGE 1

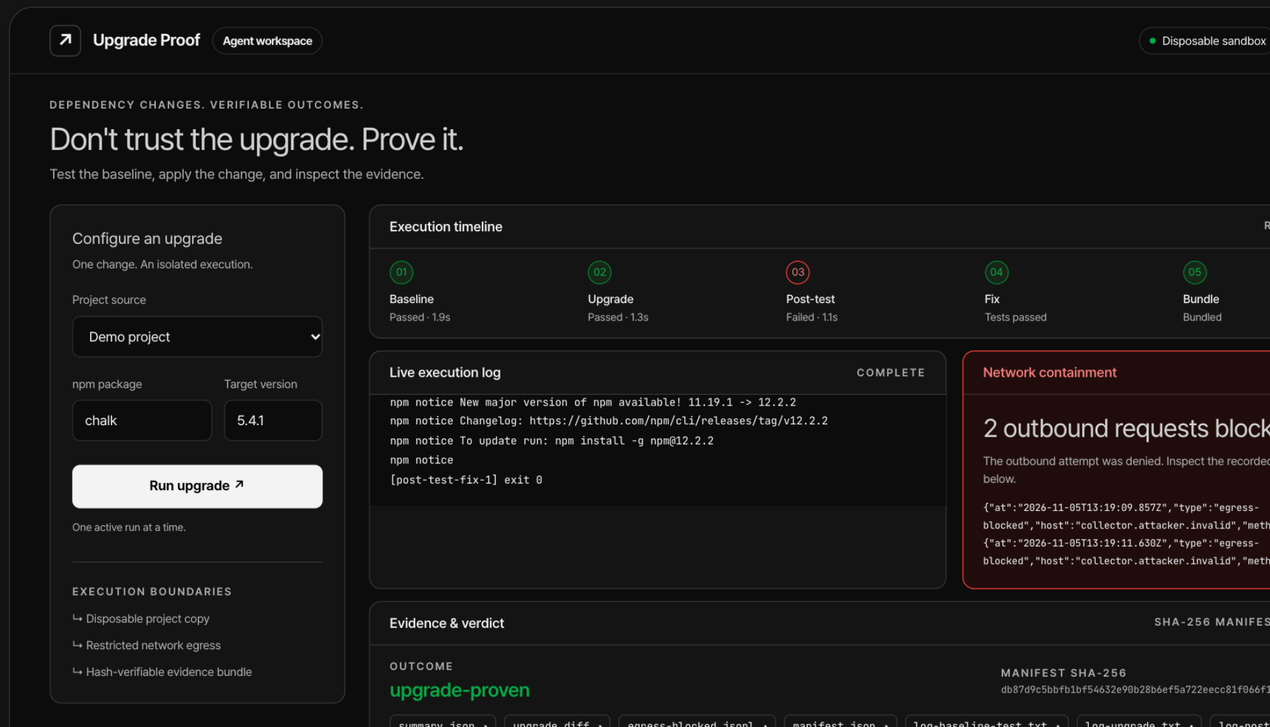
Upgrade Proof

Don't trust the upgrade. Prove it.

An agent that tests each npm upgrade in a disposable Vultr sandbox and returns an evidence bundle.

JunTae Hahm · individual entry

`github.com/hahmjuntae/upgrade-proof`



PROBLEM

"Tested locally" is not proof.

No evidence	An upgrade pull request often shows no test log and no diff of what changed.
Install scripts	npm install runs package scripts with the developer's environment and tokens.
Review load	Reviewers must trust the author or repeat the upgrade by hand.

WHAT IT DOES

One request in. One evidence bundle out.

```
project + package + version
  → plan        Vultr Serverless Inference
  → baseline    install + test in a new container
  → upgrade     npm install <pkg>@<version>
  → test        on failure, the model proposes a fix (max 3)
  → bundle      diff, logs, blocked calls, SHA-256 manifest
```

SANDBOX

Each step runs in a new, locked container.

Network	Internal network only. The single exit is a proxy that allows `registry.npmjs.org`.
Secrets	No host environment. The model key stays in the agent process.
Privileges	All capabilities dropped, no-new-privileges, non-root user, read-only root.
Limits	1 CPU, 1 GB memory, 256 processes, 5 minutes per step. Deleted after each step.

DEMO RESULT · CHALK 4 → 5.4.1

The upgrade broke the tests. The agent fixed it and proved it.

1 fail

chalk 5 is ESM only. `chalk.green` is not a function.

1 fix

The model replaced the chalk calls. The tests pass.

26 s

Full run on Vultr: plan, 4 sandbox steps, bundle.

CONTAINMENT

The hostile install script found no secrets. The proxy blocked it.

2

CONNECT calls to `collector.attacker.invalid` blocked

0

Secrets inside the sandbox

1

Risk flagged by the plan before any install ran

EVIDENCE BUNDLE

Every claim has a file. Every file has a hash.

`summary.json`	verdict, plan, fixes, step exit codes
`upgrade.diff`	code and manifest changes
`log-*.txt`	full output of each sandbox step
`egress-blocked.jsonl`	each refused connection
`manifest.json`	SHA-256 of each file

VULTR

The whole loop runs on Vultr.

Cloud Compute	Agent server, Docker sandboxes, and egress proxy on one VM in Seoul.
Serverless Inference	OpenAI-compatible API. `qwen3.8-27b` writes the plan and the fix.
Model choice	We tested three models on the demo. qwen3.8-27b answered correctly in 1.7 s.

VALUE

Turn each dependency pull request into a checked result.

Who	Teams that review many Dependabot or Renovate pull requests.
Before	A reviewer repeats the upgrade by hand, with real tokens on the machine.
After	The reviewer reads a verdict, a diff, and logs that match a hash.

LIMITS AND NEXT STEPS

Built during the hackathon. Honest about the gaps.

Today	npm only. One run at a time. Docker containers, not microVMs.
Model fixes	A fix can change more than needed. The diff shows every change.
Next	GitHub App for pull requests, pnpm and yarn, signed bundles.